LING2136/4136

Advanced Statistical Methods for Language Students

Session-11: Inference Based on ROPEs

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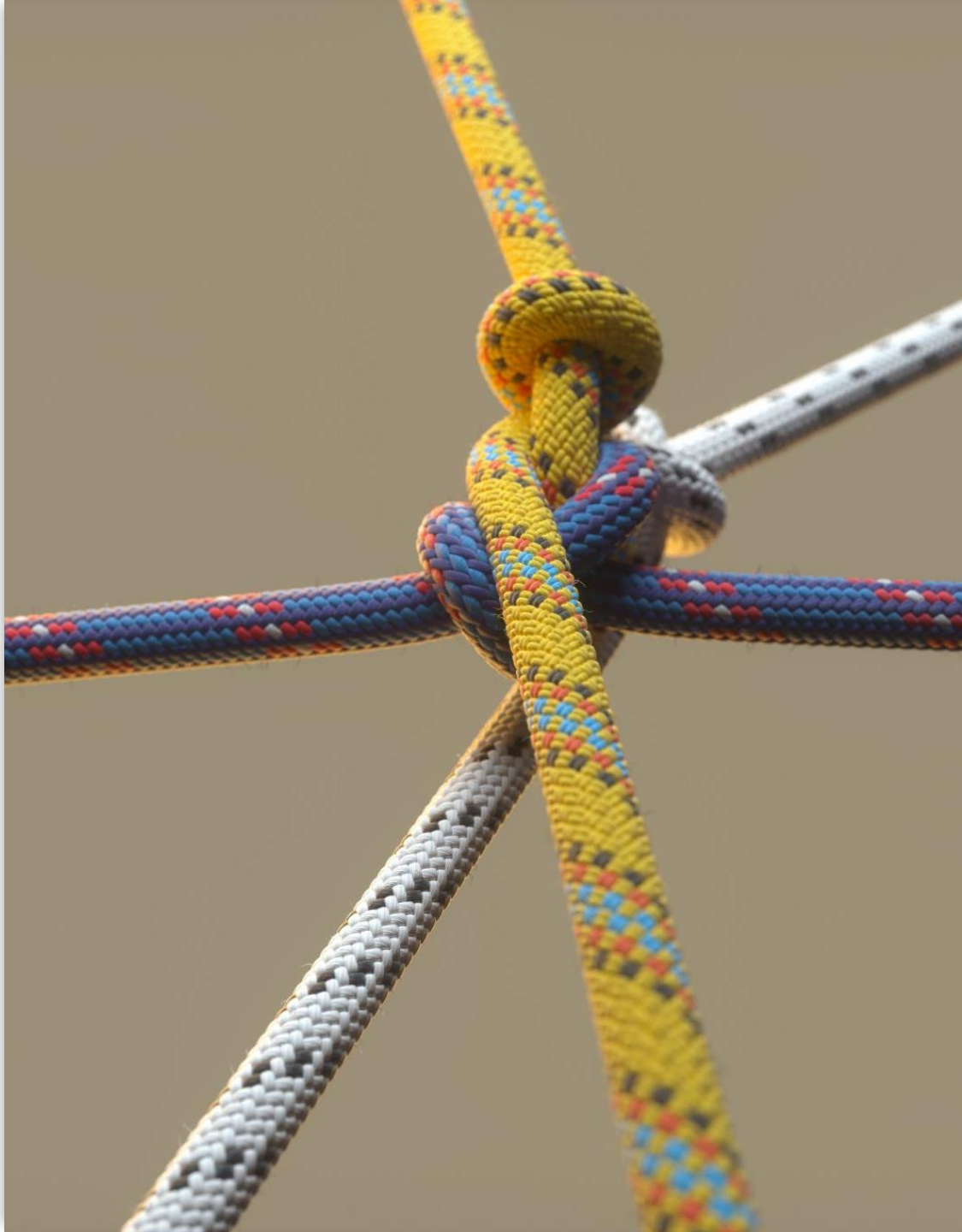
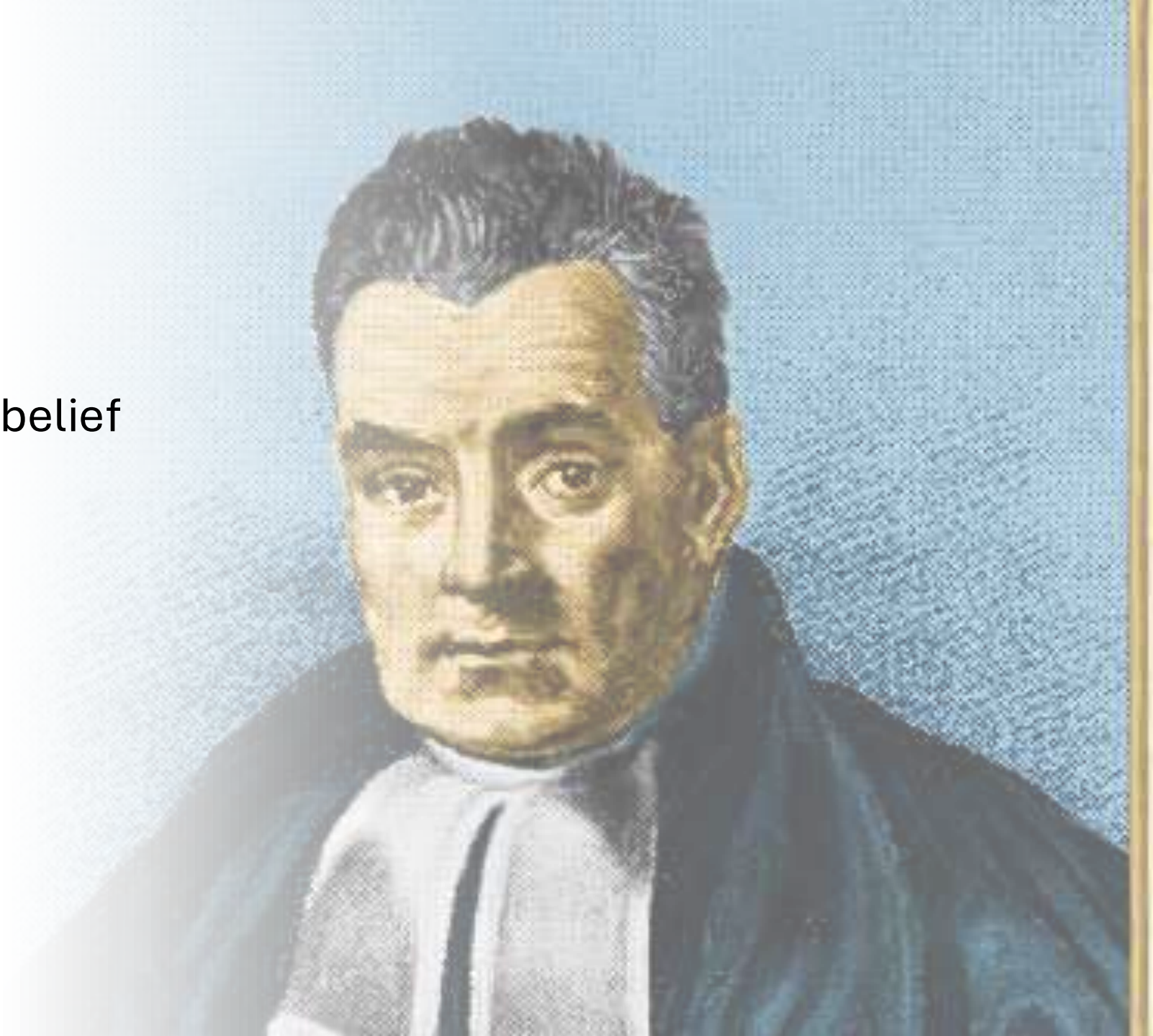
Seminar 11: Inference based on ROPEs

Seminar overview

- Quick recap on priors, posteriors, Bayesian in general
- ROPEs
  - Recap tutorial
  - In-class exercise
  - Wrap-up



belief → evidence → belief

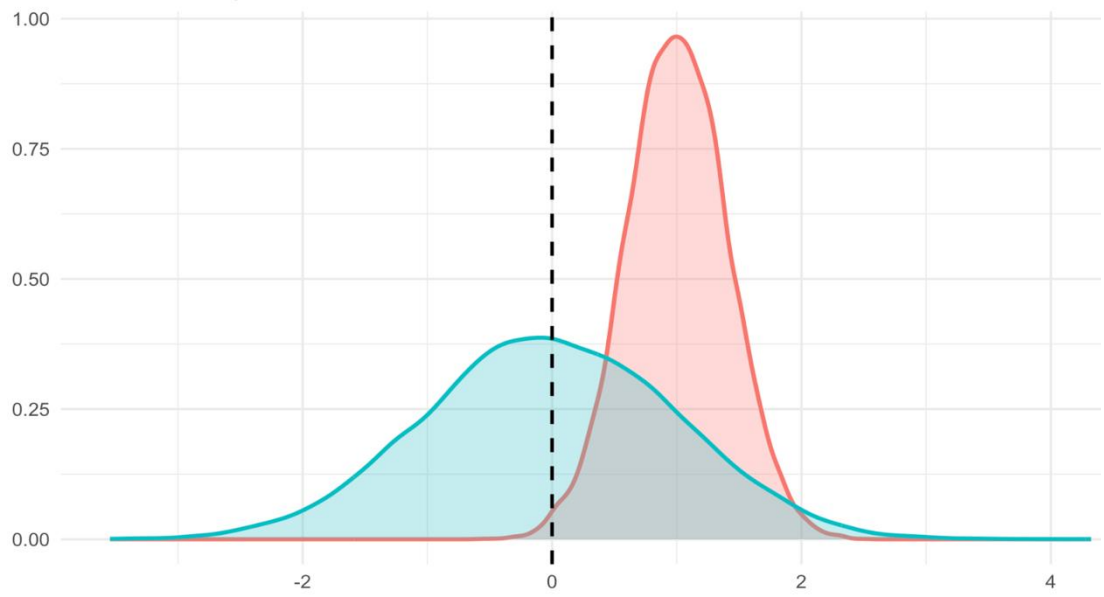


Bayesian inference for hypothesis testing

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

Bayesian inference – evaluate model

- Describe uncertainty: credible intervals
- Probability of direction
- Bayes factors



Testing hypotheses using Bayes factors

$$\frac{P(D \mid H_1)}{P(D \mid H_0)}$$

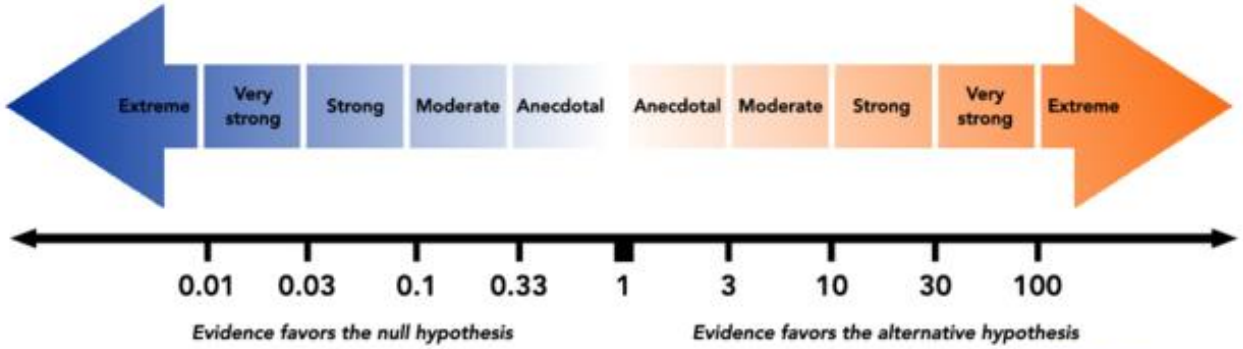
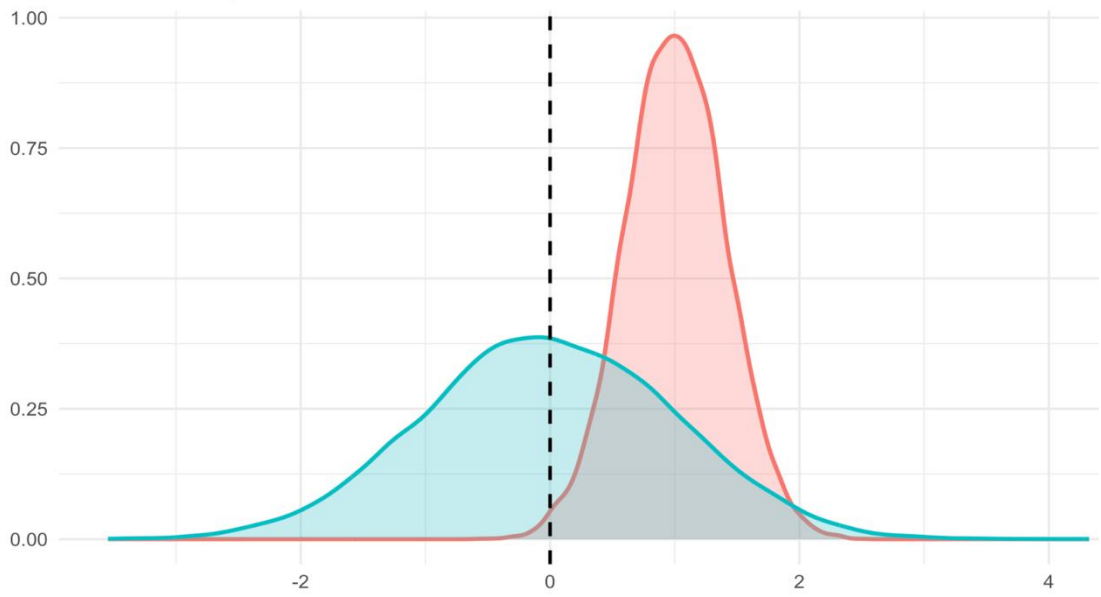


Fig. 1 Lee and Wagenmakers' classification scheme for interpreting Bayes factors (BF_{10}). This classification scheme [18, 23], which has been adopted in JASP, provides descriptive labels for interpreting a range of Bayes factors. While this scheme provides a useful starting point for understanding Bayes factor values and may be suitable for many research questions, Bayes factors should be carefully interpreted in light of the research question at hand

Testing hypotheses using Bayes factors

$$\frac{P(\theta^* = x \mid D, M_1)}{P(\theta^* = x \mid M_1)}$$



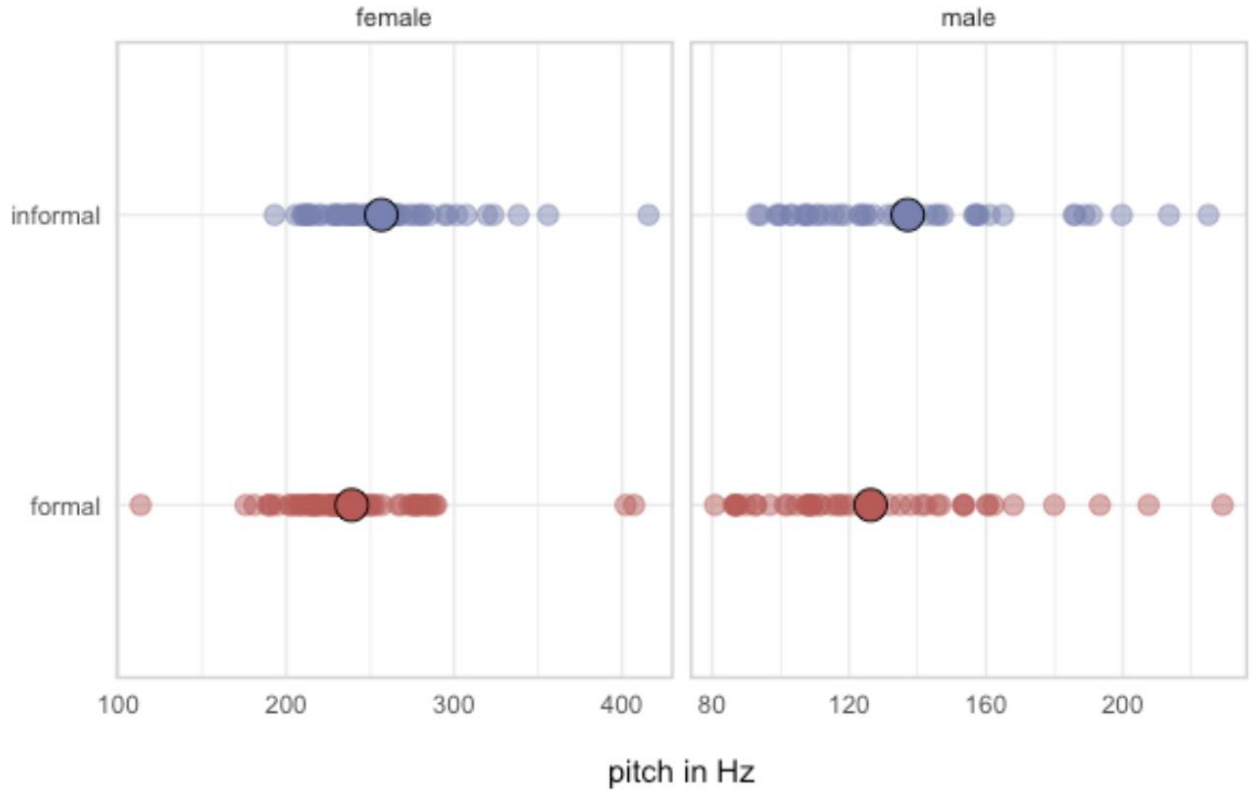
NHST → Bayes factors → ROPEs

Can we reject the null?

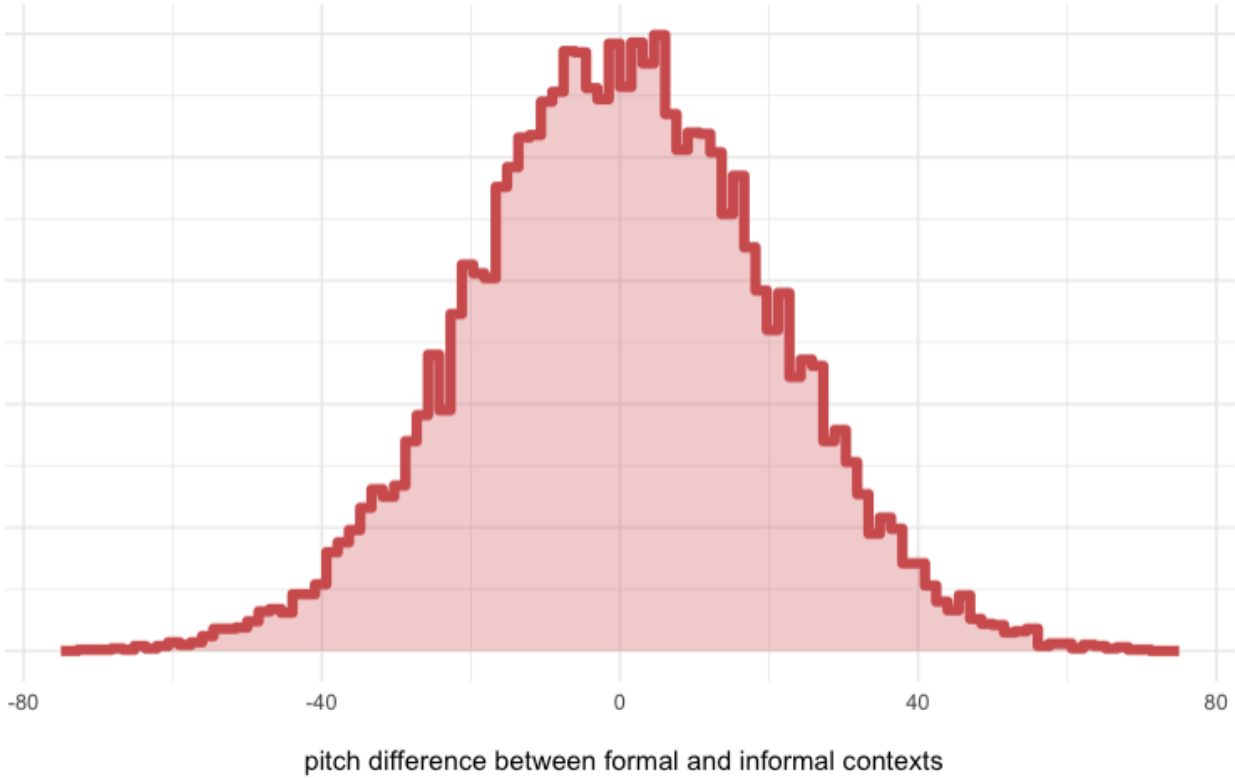
Can we reject the null OR get evidence in favor of the null?

Can we test meaningful hypotheses?

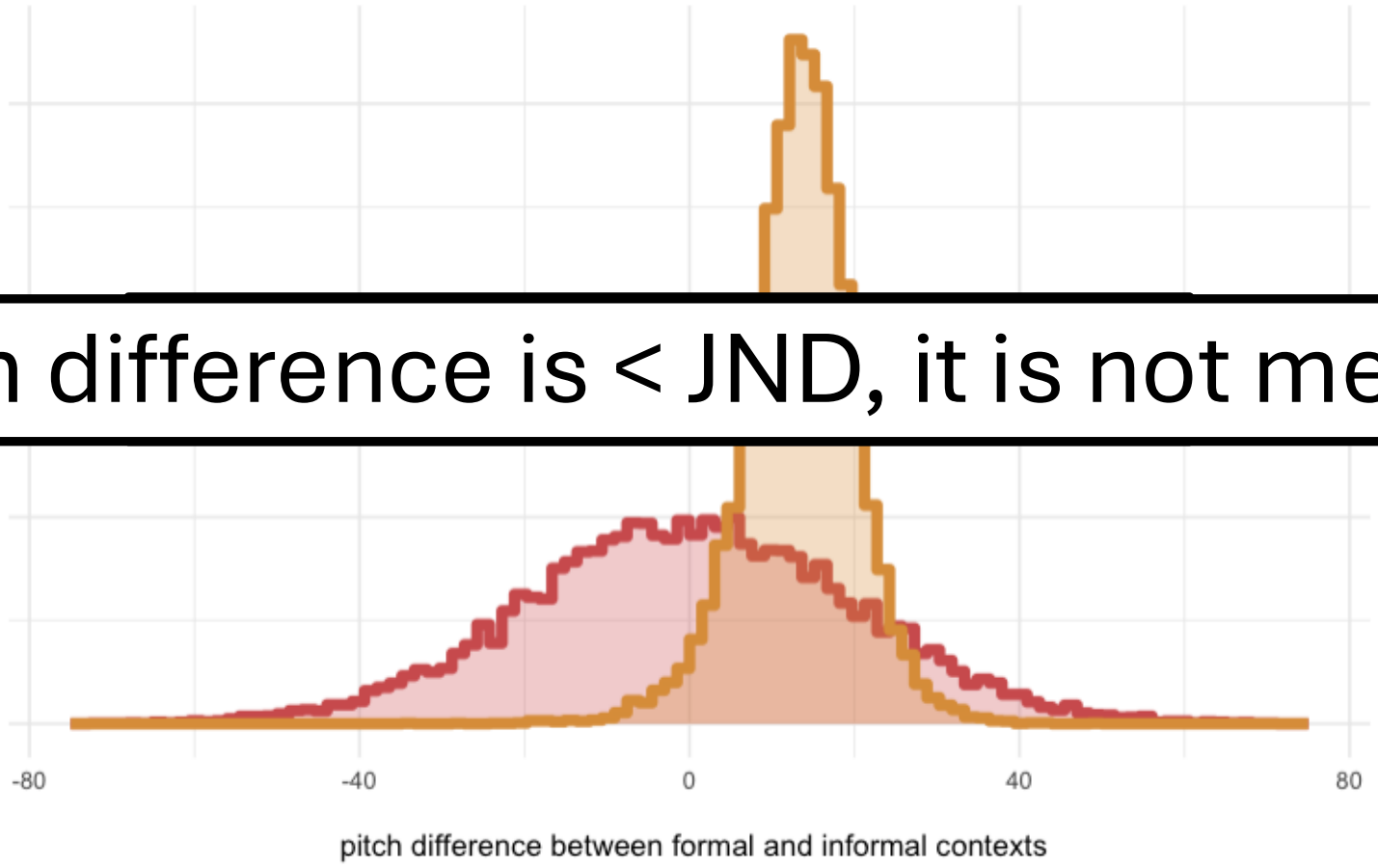
Voice pitch ~ social context




```
formula <- bf(pitch ~ context +
                      gender +
                      (1 + context | subject) +
                      (1 + context | sentence))
```


-80
-40
0
40
80
pitch difference between formal and informal contexts

If a pitch difference is < JND, it is not meaningful

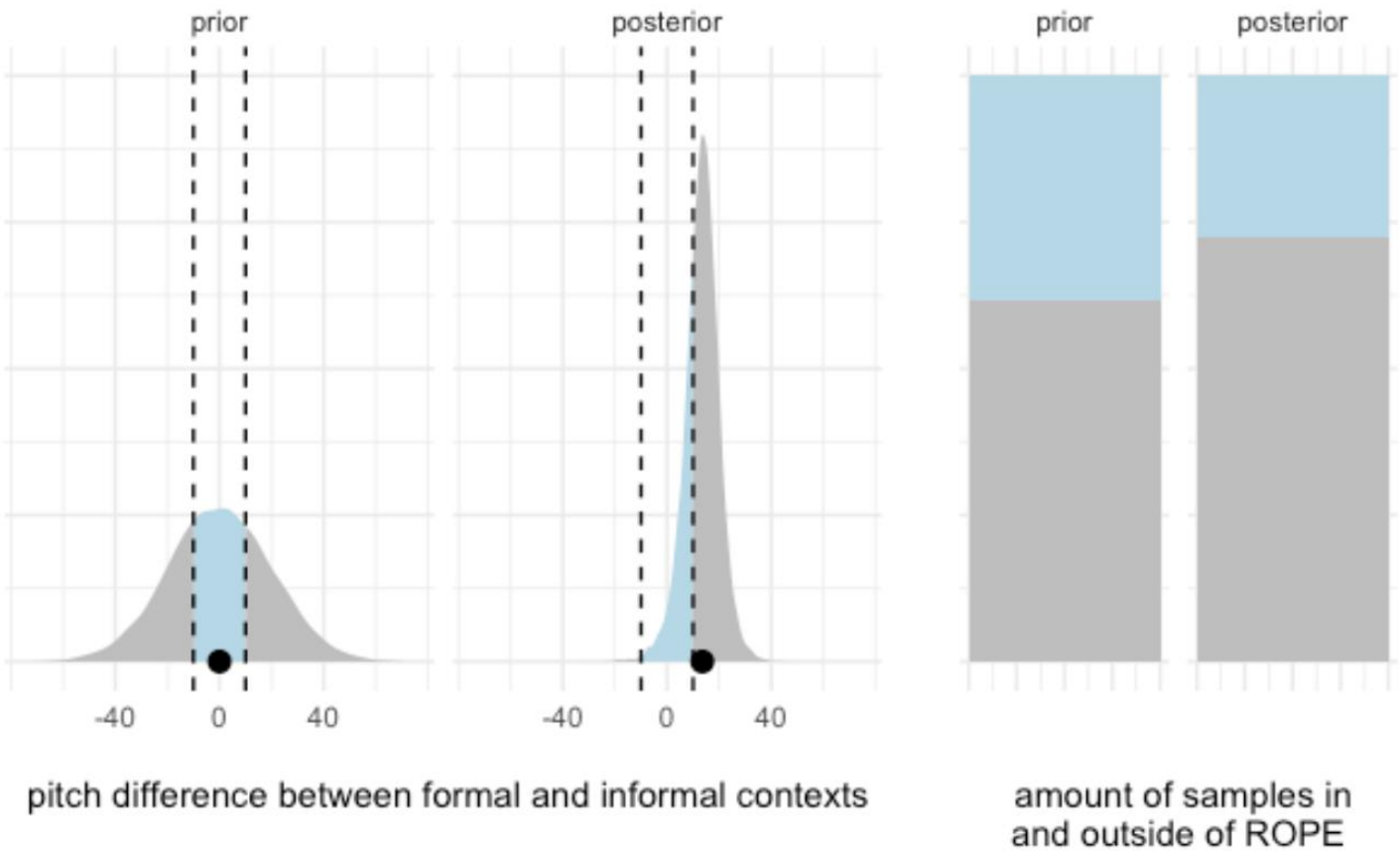


Bayes factors for interval-based hypotheses

$$\frac{P(D \mid H_1)}{P(D \mid H_0)}$$

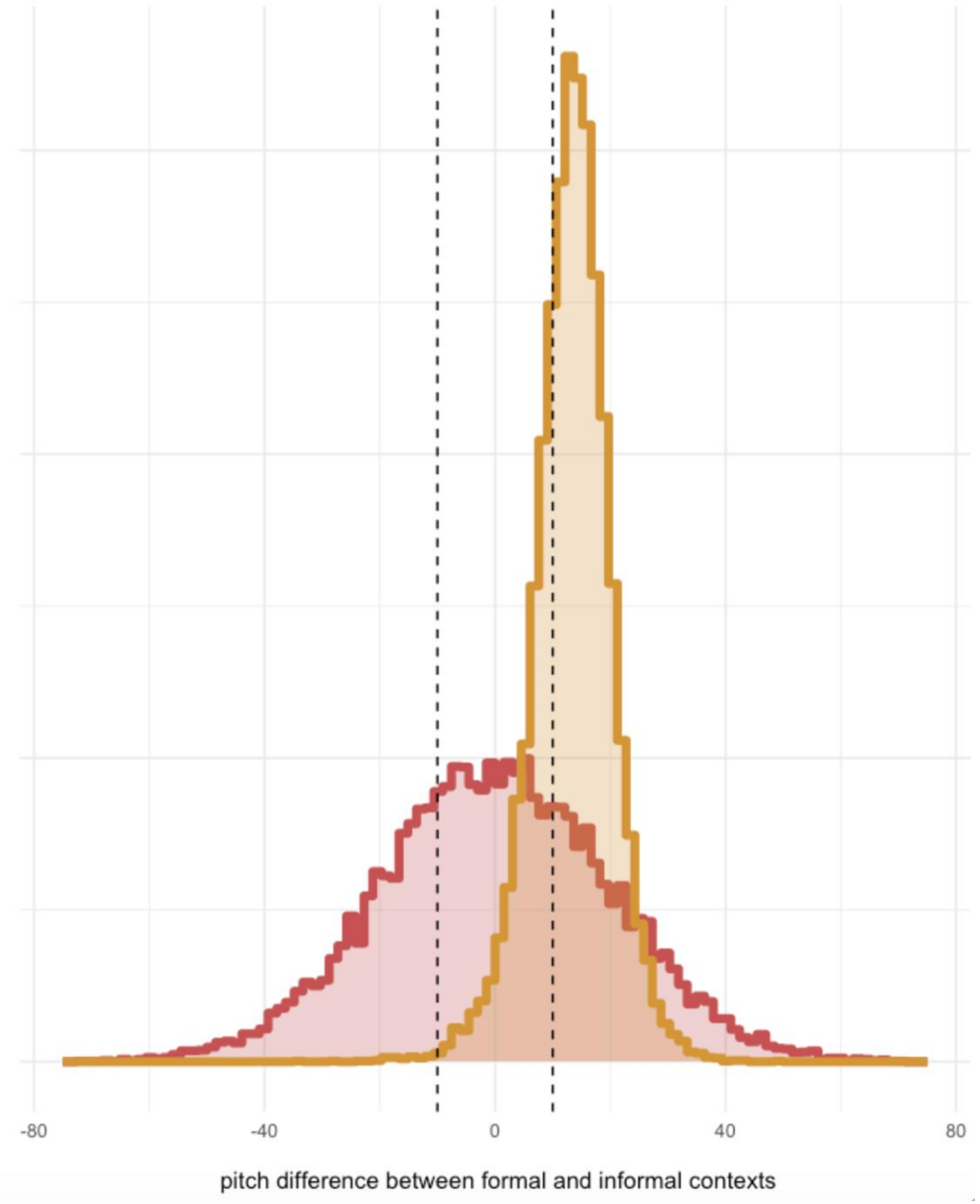
$$\frac{P(\theta \in I_0 \mid D, M_e)}{P(\theta \in I_1 \mid D, M_e)} \frac{P(\theta \in I_0 \mid M_e)}{P(\theta \in I_1 \mid M_e)}$$

Bayes factors for interval-based hypotheses



Bayes factors for interval-based hypotheses

```
BF_1 <- bayesfactor_rope(posterior = fit,
                         prior = fit_prior,
                         null = rope,
                         parameter = "b_context1")
```

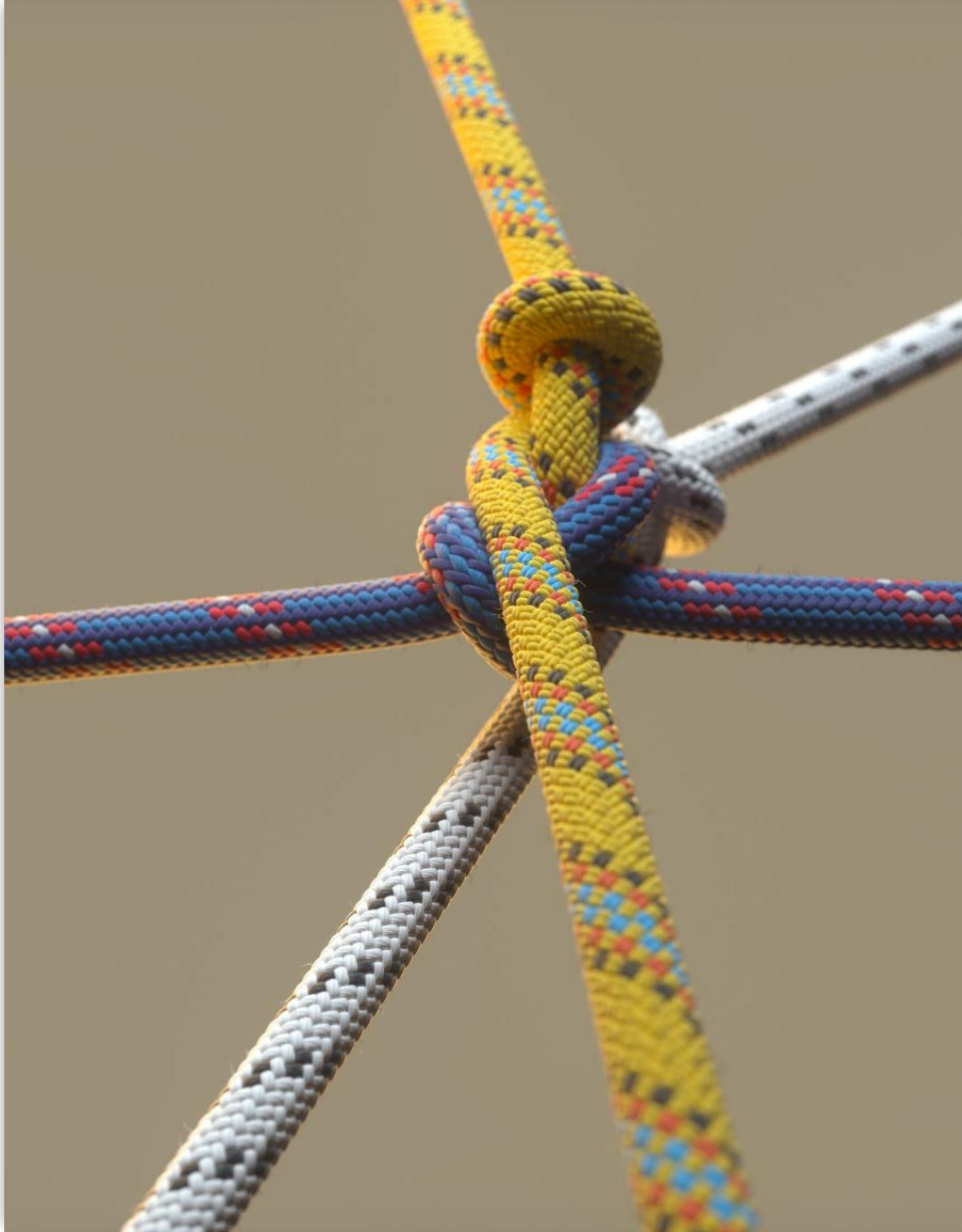


ROPEs and SESOIs

- Is the effect big enough to matter?

- Theoretical relevance
- Measurement precision
- Domain knowledge

- Reset and re-evaluate priors and ROPEs



(A) Bayes factor in favour of alternative

from moderate evidence against the null
to very strong evidence for the null

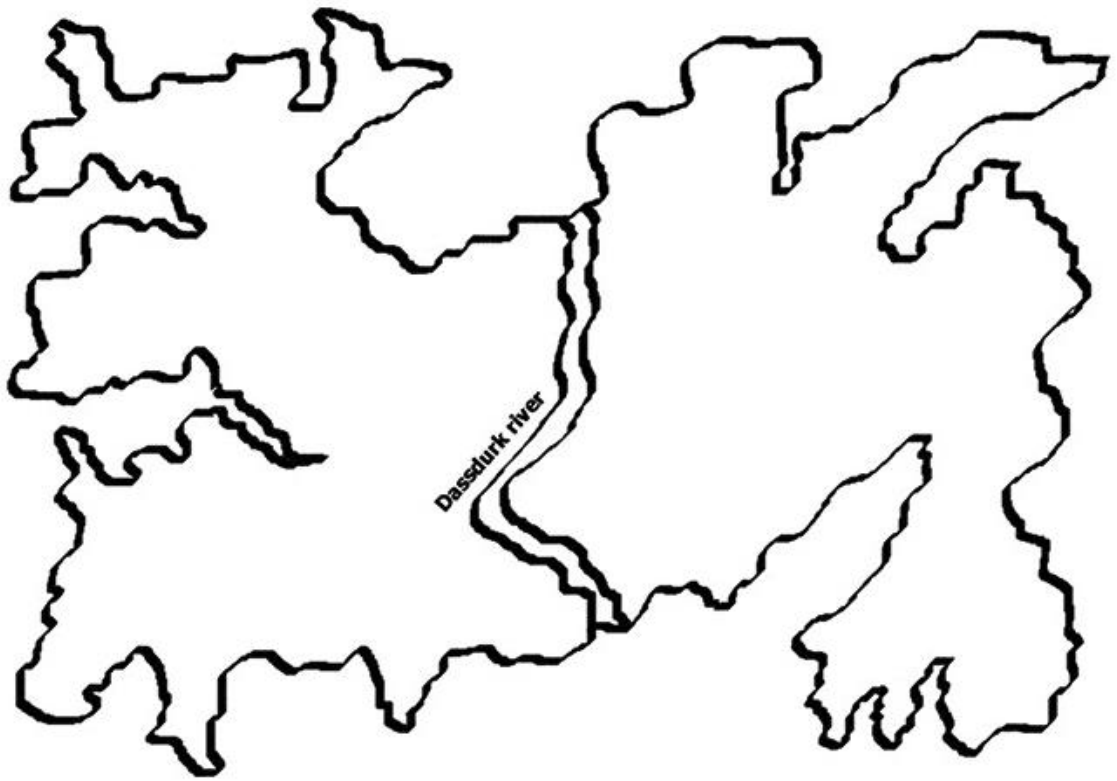
ROPE in Hz centered on 0 by 2 Hz steps	10	15	20	25	30
[0.3, -0.3]	3.1	2.89	2.61	2.21	2.25
[1, -1]	3.15	3.03	2.71	2.25	2.21
[2, -2]	3.15	2.99	2.63	2.24	2.23
[4, -4]	3.26	2.94	2.54	2.13	2.08
[6, -6]	3.17	2.77	2.25	1.85	1.81
[8, -8]	2.83	2.45	1.94	1.58	1.42
[10, -10]	2.69	2.07	1.61	1.33	1.08
[12, -12]	2.31	1.63	1.26	0.99	0.82
[14, -14]	1.88	1.23	0.96	0.75	0.62
[16, -16]	1.65	0.92	0.74	0.55	0.45
[18, -18]	1.18	0.72	0.49	0.38	0.32
[20, -20]	0.89	0.48	0.36	0.27	0.23
[22, -22]	0.6	0.34	0.24	0.19	0.15
[24, -24]	0.39	0.23	0.16	0.12	0.1
[26, -26]	0.24	0.16	0.11	0.08	0.06
[28, -28]	0.18	0.13	0.07	0.06	0.04
[30, -30]	0.16	0.09	0.04	0.04	0.03
[32, -32]	0.14	0.06	0.03	0.03	0.02
[34, -34]	0.1	0.05	0.02	0.02	0.01
[36, -36]	0.05	0.04	0.01	0.02	0.01
[38, -38]	0.02	0.03	0.01	0.01	0.01
[40, -40]	0.05	0.02	0.01	0.01	0

prior standard deviation in Hz

Bayes factor

- 10 < strong for H1
- 3 < moderate for H1
- 1 = inconclusive
- 0.33 > moderate for H0
- 0.1 > strong for H0
- 0.03 > very strong for H0

Exercise



Winter & Matlock, 2013